

Digital Buck-Boost (LTC3440): Working Document

Digital Audio Player • Rev 5

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Contents

1	Summary	1
2	Design Requirements	1
3	Pin Configuration	1
4	External Components	2
5	Equations and Results	2
6	Important Values	3
7	Open Problems	3
8	Revision Log	4
9	Sources	4

1. Summary

Field	Value
Manufacturer / MPN	Analog Devices, LTC3440EMS#TRPBF
Package	10-lead MSOP (MS)
Function	Synchronous buck-boost, SYS → +3.3 V digital rail

2. Design Requirements

Requirement	Value
Input	SYS, 3.0 to 4.2 V
Output	3.3 V
Load	Digital branch: ESP32-S3, microSD, display logic, DAC digital
Output current	300 mA max ($\approx$ 253 mA S4 peak)
Switching frequency	600 kHz (0.48 to 0.72 MHz), separated from the analog LTC3440 (0.96 to 1.44 MHz)
Inductor ripple target	30%
Mode	Fixed frequency (MODE/SYNC to GND)
Constraints	Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1	RT	R_T to GND
2	MODE/SYNC	GND (fixed frequency)
3	SW1	L
4	SW2	L
5	GND	GND
6	VOUT	+3V3, C_{OUT} , R1
7	VIN	SYS, C_{IN}
8	SHDN/SS	VIN (always on, per the typical application)
9	FB	R1 / R2 midpoint
10	VC	C_{P1} and R_Z to FB

4. External Components

Ref	Value / Rating	MPN	AL-PN
L	18 μ H $\pm$ 20%; I_{sat} 1.8 / 2.8 / 3.7 A (10 / 20 / 30% drop); I_{rms} 3.3 A; 62.5 m Ω max; shielded, 5.48 $\times$ 5.28 $\times$ 5.10 mm	Coilcraft XGL5050-183MEC	#1-094
C_{IN}	4.7 μ F 16 V X7R 0805	Murata GRM21BR71C475KE51L	#1-071
C_{OUT}	22 μ F 25 V X7S 1210	Murata GCM32EC71E226KE36L	#1-073
R1	172 k Ω 0.1% 0402 (E192)	TBD	
R2	100 k Ω 0.1% 0603	YAGEO RT0603BRD07100KL	#1-066
R_T	100 k Ω 0.1% 0603	YAGEO RT0603BRD07100KL	#1-066
R_Z	15 k Ω 0.1% 0402	YAGEO RT0402BRD0715KL	#1-078
C_{P1}	1.5 nF $\pm$ 2% 50 V C0G 0402	Murata GRM1555C1H152GA01D	#1-077

5. Equations and Results

Switching frequency

$$R_T = \frac{6 \times 10^{10}}{f} = \frac{6 \times 10^{10}}{600 \times 10^3} = 100 \text{ k}\Omega \quad (0.48 \text{ to } 0.72 \text{ MHz at } \pm 20\%)$$

Source: [LTC3440 Product Page and Datasheet \(Analog Devices\)](#), oscillator.

Output voltage

$$V_{OUT} = 1.22 \left(1 + \frac{R_1}{R_2} \right) \Rightarrow R_1 = 100 \text{ k}\Omega \left(\frac{3.3}{1.22} - 1 \right) = 170.5 \text{ k}\Omega, \quad V_{OUT} = 1.22 \left(1 + \frac{172}{100} \right) = 3.318 \text{ V}$$

Source: [LTC3440 Product Page and Datasheet \(Analog Devices\)](#), output voltage.

Inductor (f in MHz, 300 mA, 30% ripple)

$$L_{\text{boost}} > \frac{V_{\text{IN(MIN)}} (V_{\text{OUT}} - V_{\text{IN(MIN)}})}{f I_{\text{OUT(MAX)}} \text{Ripple } V_{\text{OUT}}} = \frac{3.0 \times 0.3}{0.6 \times 0.3 \times 0.3 \times 3.3} = 5.05 \mu\text{H}$$
$$L_{\text{buck}} > \frac{V_{\text{OUT}} (V_{\text{IN(MAX)}} - V_{\text{OUT}})}{f I_{\text{OUT(MAX)}} \text{Ripple } V_{\text{IN(MAX)}}} = \frac{3.3 \times 0.9}{0.6 \times 0.3 \times 0.3 \times 4.2} = 13.1 \mu\text{H} \rightarrow 18 \mu\text{H}$$

Source: [LTC3440 Product Page and Datasheet \(Analog Devices\)](#), inductor selection.

Ripple with 18 μH and peak inductor current (3.0 V in, $\eta \approx 0.85$)

$$\text{Ripple}_{\text{buck}} = 30\% \times \frac{13.1}{18} = 21.8\% (65 \text{ mA}_{\text{pp}}), \quad \text{Ripple}_{\text{boost}} = 30\% \times \frac{5.05}{18} = 8.4\% (25 \text{ mA}_{\text{pp}})$$
$$I_{L,\text{pk}} = \frac{0.3 \times 3.3}{3.0 \times 0.85} + \frac{0.025}{2} = 0.40 \text{ A} \quad (4.2 \text{ V in} : 0.33 \text{ A})$$

Copper loss, output capacitance (0.5% ripple) and gate-charge current (f in MHz)

$$P_{\text{DCR}} = I_L^2 R_{\text{DCR}} = 0.388^2 \times 0.0625 = 9.4 \text{ mW}$$
$$C_{\text{OUT}} \geq \frac{I_{\text{OUT}} (V_{\text{OUT}} - V_{\text{IN(MIN)}}) \cdot 100}{\% \text{Ripple} \cdot V_{\text{OUT}}^2 \cdot f} = \frac{0.3 \times 0.3 \times 100}{0.5 \times 3.3^2 \times 0.6 \times 10^6} = 2.75 \mu\text{F} \quad (22 \mu\text{F fitted})$$
$$I_Q = f (0.75 V_{\text{IN}} + 0.25 V_{\text{OUT}}) = 0.6 \times (0.75 \times 4.2 + 0.25 \times 3.3) = 2.385 \text{ mA}$$

Source: [LTC3440 Product Page and Datasheet \(Analog Devices\)](#).

Four-switch region and boost right-half-plane zero (3.0 V in, 300 mA, 18 μH)

$$V_{\text{IN}} = V_{\text{OUT}} (1 - 150 \text{ ns} \cdot f) \text{ to } \frac{V_{\text{OUT}}}{1 - 150 \text{ ns} \cdot f} = 3.00 \text{ to } 3.63 \text{ V}$$
$$f_{\text{RHPZ}} = \frac{V_{\text{IN}}^2}{2\pi I_{\text{OUT}} L V_{\text{OUT}}} = \frac{3.0^2}{2\pi \times 0.3 \times 18 \mu\text{H} \times 3.3} = 80 \text{ kHz}$$

Source: [LTC3440 Product Page and Datasheet \(Analog Devices\)](#).

6. Important Values

Item	Value
Switch current limit	2.7 A typ (2.5 to 2.9 A)
Output	3.318 V with 172 k Ω
Inductor ripple	21.8% buck / 8.4% boost with 18 μH
Peak inductor current	0.40 A at 300 mA, 3.0 V in; well below the 1.8 A 10%-drop point
Inductor at switch limit	2.9 A max $\rightarrow \approx 21\%$ inductance drop; fault only, soft saturation
Gate-charge I_Q	2.385 mA at 4.2 V in ($\approx 10 \text{ mW}$)
Boost RHP zero	$\approx 80 \text{ kHz}$; loop crossover must stay well below

7. Open Problems

None.

8. Revision Log

Rev	Date	Change
0	2026-09-23	Initial issue: 600 kHz, 3.3 V / 300 mA, inductor, output capacitance, quiescent current.
1	2026-09-23	MODE/SYNC to GND; R1 172 k Ω .
2	2026-09-23	Inductor Coilcraft XGL5050-183MEC.
3	2026-09-23	Inductor I_{sat} by drop column recorded.
4	2026-09-23	Inductor library part assigned.
5	2026-09-25	Reformatted; linked contents and sources; open problems closed.

9. Sources

1. [LTC3440 Product Page and Datasheet \(Analog Devices\)](#)